

Class 12: Wi-Fi Sensing

Angle from the Antennas, Distance from the Subcarriers

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CIS 4930/5930 Future Edge Networks

xinliulab.github.io/FSU-CIS4930-CIS5930-Future-Edge-Networks

What your access point knows without asking

Where is the phone?

to 40 cm, median
(SpotFi, 2015)

Is someone home?

presence, motion,
gestures — no camera

Are they breathing?

breathing, heartbeat
(2FiA — Class 13)

Q: Which extra sensor did the access point need for all of this?

A: None. Every answer was read out of the **channel estimates** the AP already computes just to decode packets. This lecture is about how.

Kotaru et al., SpotFi, SIGCOMM 2015 · 2FiA appears in Paper Practicum 5 (Class 13)

RSSI is a light meter, phase is a ruler

RSSI — signal strength

- One number per packet.
- All paths, summed as power.
- Swings dBs if you just stand up.
- Good for: rough presence.

Phase — of the channel H

- One per subcarrier, per antenna.
- Turns 360° every wavelength.
- At 5 GHz: $\lambda = 6$ cm.
- 1° of phase ≈ 0.17 mm of path.

Wi-Fi sensing is the move from reading power to reading **phase** — from a light meter to a ruler.

Adib, MIT MAS.S60, Lecture 2: RF localization — the RSSI-to-phase hierarchy

The receiver must learn H before bit one

Every packet opens with training symbols the receiver already knows (Class 8).

On each subcarrier f and each antenna, the receiver solves

$$y = H \cdot x_{\text{known}} \Rightarrow H = y/x_{\text{known}}$$

H is one **complex number**: $|H|$ says how much the wave faded, $\arg H$ says how far it travelled — for a single path of length r :

$$H = \alpha e^{-j2\pi f r/c} \quad (\text{one turn of phase per wavelength of } r)$$

Q: What does the modem do with H after equalizing the packet?

A: Throws it away — thousands of times per second. CSI tools simply keep it. Sensing is recycling.

CSI = Channel State Information: the collection of these H values c = speed of light

Ninety numbers per packet, for free

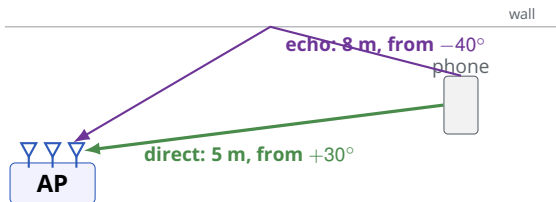


- The classic Intel 5300 tool exports 3 antennas $\times$ 30 subcarriers = **90 complex values per packet**.
- Hundreds of packets per second $\Rightarrow$ a movie of the channel: $H[\text{time}, \text{antenna}, \text{subcarrier}]$.

Phase twists exaggerated on the frequency axis so they are visible.

Halperin et al., Linux 802.11n CSI Tool · Tsinghua Wireless Sensing Tutorial, "Understanding CSI"

The room writes itself into the matrix

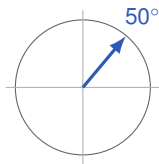


Each path ℓ stamps the whole matrix at once:

$$H[m, k] = \sum_{\ell} \alpha_{\ell} \underbrace{e^{-j2\pi f_k \tau_{\ell}}}_{\text{delay } \tau_{\ell}} \underbrace{e^{-j\frac{2\pi}{\lambda} m d \sin \theta_{\ell}}}_{\text{angle } \theta_{\ell}}$$

The AP receives only the **sum**. Sensing is un-summing: recover each path's $(\theta_{\ell}, \tau_{\ell})$ from the matrix.

The catch: every phase arrives wrapped



what you measure:
the needle, not the turns

Our 5 m direct path at $f = 5.2$ GHz has turned

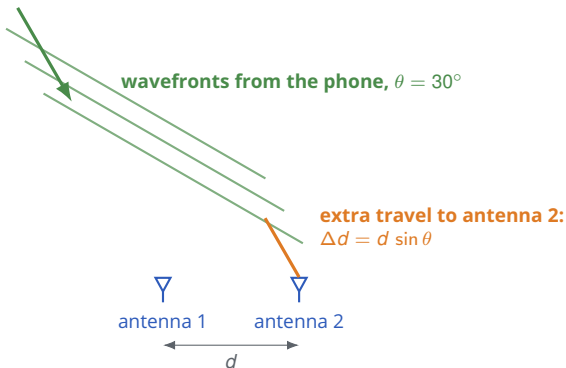
$$f \cdot \tau \approx 5.2 \text{ GHz} \times 17 \text{ ns} \approx \mathbf{87 \text{ full turns.}}$$

The receiver sees only the leftover fraction — like an odometer showing nothing but its last digit.

One phase reading says: “the path is $x + n \cdot 6$ cm, for some integer n .” Useless alone.

The fix is never one reading — it is a **difference** along an axis:
across **antennas** → **angle**; across **subcarriers** → **distance**.

Two ears, one voice



Q: The wavefront is a straight line. Why does antenna 2 hear it late?

A: Because the line is tilted by the arrival angle θ . The lag **is** the direction — your ears run this algorithm all day.

The phase difference is the angle

$$\Delta\varphi = \frac{2\pi}{\lambda} d \sin \theta$$

extra path $d \sin \theta$, charged at one turn per wavelength

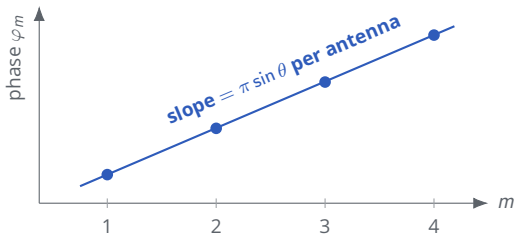
Running example: $\lambda = 6 \text{ cm}$, $d = 3 \text{ cm}$, phone at $\theta = 30^\circ$

$$\Delta\varphi = \frac{2\pi}{6 \text{ cm}} \cdot 3 \text{ cm} \cdot \sin 30^\circ = \pi/2 \text{ — a quarter turn per antenna}$$

Q: Why do arrays space antennas exactly $d = \lambda/2$?

A: Wrapping, again. At $d = \lambda/2$, $\Delta\varphi = \pi \sin \theta$ stays inside $(-\pi, \pi]$: every angle gets its own phase step. Space them wider and two directions share one reading.

M antennas: the phases form a line

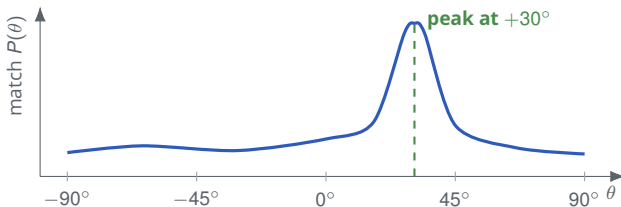


- $\varphi_m = m \cdot \pi \sin \theta$: the direction sets the **slope** across the array.
- The pattern a candidate angle would produce is its **steering vector** $a(\theta)$.
- Finding θ = finding which template line fits the measured column of H .

Boresight ($\theta = 0$): slope 0 — all antennas in step. Endfire ($\theta = 90^\circ$): half a turn each.

Scanning every angle in software

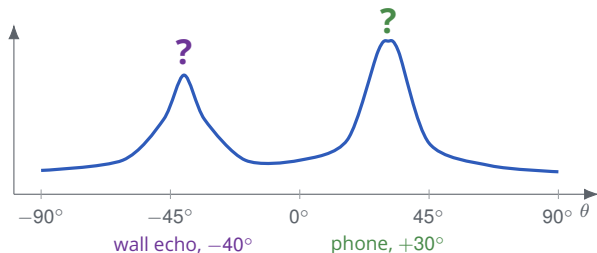
For each candidate θ from -90° to $+90^\circ$: how well does $a(\theta)$ match the measured antenna column?



This is a phased-array sweep with no moving parts:
the beam is a `for`-loop over θ , run after reception.

Xiong and Jamieson, ArrayTrack, NSDI 2013

Multipath grows extra peaks

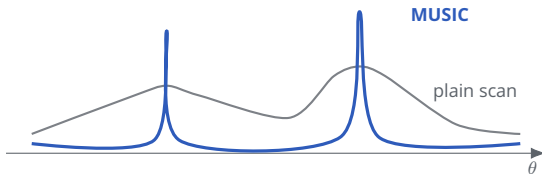


- Every arrival is real: the echo has its own slope, its own peak.
- On this axis alone, **nothing says which peak is the phone.**
- ArrayTrack's brute-force help: more antennas $\rightarrow$ sharper peaks (16 antennas $\rightarrow$ 23 cm median).

We need a second ruler that tells the two peaks apart.
Hold that thought — first, let us sharpen this one.

MUSIC, in one slide

- Collect many packets $\rightarrow$ average $\rightarrow$ a covariance of the antenna readings.
- Its eigenvectors split cleanly: a few span the **signal** (one per path), the rest span pure **noise**.
- A true arrival angle's steering vector is orthogonal to the noise part.
- So plot $1/(\text{projection onto noise})$: division by almost-zero $\rightarrow$ needle-sharp spikes.



Schmidt, IEEE Trans. AP 1986 · Have a try: Princeton COS463 MUSIC lab; Tsinghua tutorial [naive_aoa](#)

Same trick: rotate the ruler

Fix one antenna. Walk along the **frequency axis** instead:

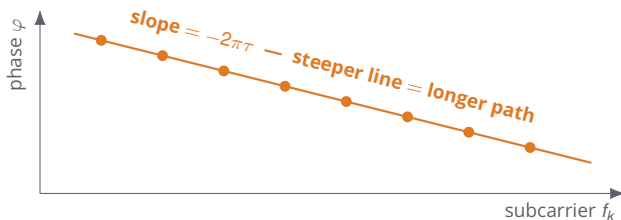
$$\varphi(f) = -2\pi f \tau \quad (\tau = \text{time of flight} = r/c)$$

- One frequency alone is hopeless: our 5 m path is **87 turns** deep in wraps.
- But two nearby frequencies differ by only $\Delta\varphi = 2\pi \Delta f \tau$ — a tiny, unwrapped step.
- Same sentence as last section: a **difference across a small step beats wrapping**.

Q: Where do we get many nearby frequencies, measured at the same instant?

A: You have known the answer since Class 8: **OFDM**. Every Wi-Fi packet is 52+ equally spaced tones, each reporting its own phase.

Two tones make a slope

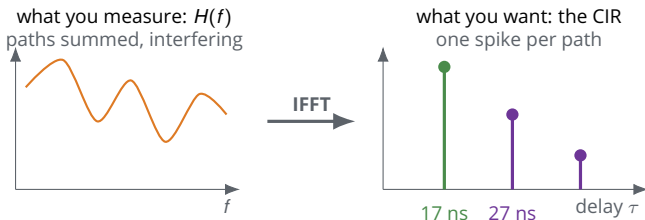


Running example, 5 m path: 1.9° per 312.5 kHz subcarrier step,
 $\approx 120^\circ$ of total fall across one 20 MHz packet.

The step $\Delta f = 312.5$ kHz wraps only beyond $c/\Delta f = \mathbf{960\text{ m}}$ — indoors, never.

Vasisht et al., Chronos, NSDI 2016 — slides 11–31 walk exactly this argument

The IFFT sorts the room by distance



H across frequency and the echoes across delay are a Fourier pair.

The IFFT is the sorting machine.

CFR = Channel Frequency Response (H vs. f) CIR = Channel Impulse Response (H vs. τ)

Resolution is bandwidth

Two CIR spikes merge unless their delays differ by $\gtrsim 1/B$:

Channel width B	tap width $1/B$	in meters	our 5 m vs. 8 m paths?
20 MHz	50 ns	15 m	merged
40 MHz	25 ns	7.5 m	merged
80 MHz	12.5 ns	3.75 m	barely
160 MHz	6.25 ns	1.9 m	split

Two escapes, and you have met both ideas today:

- **Super-resolution math** — run MUSIC along the **frequency axis** too (SpotFi, next section).
- **Buy bandwidth** — hop across all Wi-Fi bands and stitch the slopes into one giant sweep: Chronos, sub-nanosecond ToF.

Chronos stitches 2.4 and 5 GHz bands on commodity Wi-Fi cards

What the clocks ruin

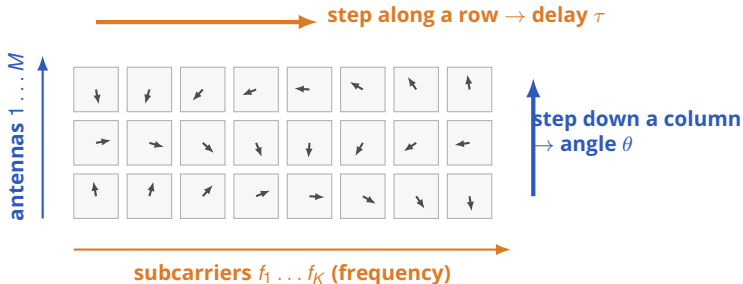
Four lies in every raw CSI capture:

- **Unsynced clocks:** TX and RX never agree on $t = 0$; absolute τ is biased.
- **Packet detection delay:** the bias changes packet to packet.
- **Carrier frequency offset:** a slow phase spin on top of everything.
- **Per-chain offsets:** each RF chain adds its own constant phase (calibrate once).

What survives untouched: differences. Path-to-path delay gaps, the slope shape across subcarriers, antenna-to-antenna steps.

Trust differences, not absolutes. (Chronos is, at heart, one long fight to win back the absolute.)

The matrix returns

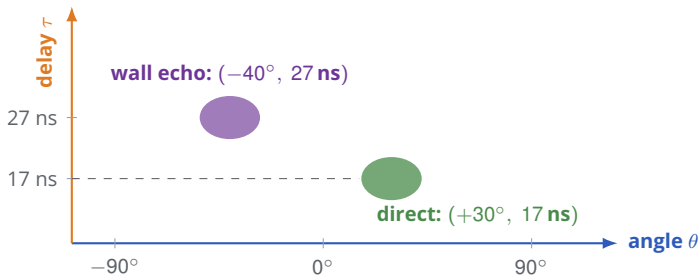


Each path twists the grid along **both** axes at once — so fit both at once: treat every (antenna, subcarrier) cell as one sensor and run MUSIC over candidate pairs (θ, τ) .

One measurement, two questions, answered jointly:
2D MUSIC over (θ, τ) .

Kotaru et al., SpotFi, SIGCOMM 2015 — Sec. 3: the 2D estimator and its smoothing trick

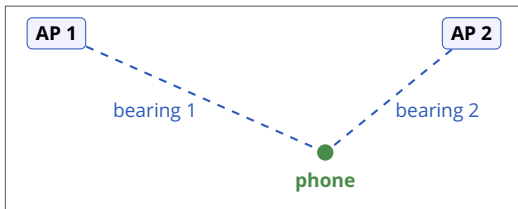
Every path is a peak in (angle, delay)



Q: Two peaks again — which one is the phone now?

A: The **earliest** one. A reflection travels farther by definition, so the direct path has the smallest τ . The delay axis breaks the tie the angle axis could not.

From (angle, delay) to a dot on the map



- Each AP contributes its direct path's **bearing**; delays down-weight APs stuck behind reflections.
- Bearings intersect → a position — no app, no extra hardware on the phone.
- SpotFi: stock 3-antenna 802.11n APs, **40 cm median error**.

That is the hook card from slide one, cashed in.

The textbook vs. the hardware

The textbook says	The hardware answers
Phase encodes distance.	...modulo 6 cm, 87 turns deep.
Subcarrier phases are clean.	A new detection delay tilts them every packet.
Antennas share one clock.	Each RF chain adds its own constant offset.
A CIR peak per path.	At 20 MHz, one tap is 15 m wide.
One target, standing still.	Chairs, doors, and people all reflect too.

Every trick in this lecture was a **difference** — slopes and steps survive what absolutes do not.

What real systems achieved

System	Pushes on	Hardware	Result
ArrayTrack '13	antenna axis	16-antenna AP	23 cm median
SpotFi '15	both axes, jointly	3-antenna AP	40 cm median
Chronos '16	frequency axis	one AP, band hops	sub-ns ToF

And the standards caught up:

- **802.11mc FTM**: round-trip ranging is in the Wi-Fi standard your phone ships with.
- **802.11bf**: a task group making sensing itself a standard Wi-Fi feature.

[ArrayTrack, NSDI'13](#) · [SpotFi, SIGCOMM'15](#) · [Chronos, NSDI'16](#) · [IEEE 802.11bf status](#)

One matrix, two axes, two rulers

Read $H[m, k] \dots$

You get

Because

down a column (antennas)

along a row (subcarriers)

both at once

Phase is a 6 cm ruler with no absolute markings.
All of Wi-Fi sensing is reading its **differences** along the matrix's axes.

One matrix, two axes, two rulers

Read $H[m, k] \dots$	You get	Because
down a column (antennas)	angle θ	$\Delta\varphi = \frac{2\pi}{\lambda} d \sin \theta$
along a row (subcarriers)	delay τ (distance)	$\Delta\varphi = 2\pi \Delta f \tau$
both at once	(θ, τ) peaks	2D MUSIC; earliest wins

Phase is a 6 cm ruler with no absolute markings.
All of Wi-Fi sensing is reading its **differences** along the matrix's axes.

Where this goes next

- **Class 13 (practicum):** CARTS runs these tricks on 5G's own uplink reference signals; 2FiA reads breathing and heartbeat out of CSI.
- **Class 14:** the AP already broadcasts compressed CSI — 802.11 beamforming feedback — and we will sense with that.

Try it yourself (both free, with code):

- Tsinghua Hands-on Wireless Sensing: [CSI features tutorial](#) — Matlab `naive_aoa / naive_tof`.
- Princeton COS463 [MUSIC lab](#) — build the AoA spectrum in NumPy.

Your phone never opted in to any of this.
Who should be allowed to run a ruler made of your Wi-Fi?